



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Guwahati
Course Structure and Syllabus

(From Academic Session 2024-25 onwards)

B.TECH
COMPUTER SCIENCE ENGINEERING

3rd SEMESTER



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY

Course Structure

(From Academic Session 2024-25 onwards)

B. Tech 3rd Semester: Computer Science Engineering

Semester III

Semester IV								
Sl. No.	Course Code	Course	Hours per Week			Credit	Marks	
			L	T	P	C	CE	ESE
Theory								
1	CS241301	Data Structure and Algorithms	3	0	0	3	30	70
2	CS241302	Object Oriented Programming	3	0	0	3	30	70
3	CS241303	Software Engineering	3	0	0	3	30	70
4	CS241304	Digital Systems	3	1	0	4	30	70
5	MA241305D	Mathematics III-D	3	0	0	3	30	70
6	HS241306	Indian Knowledge Systems	2	0	0	2	0	50
7	AC241307	Language other than MIL	1	1	0	0 PP/NP	-	50
Practical								
1	CS241311	Data Structure and Algorithms Lab	0	0	2	1	15	35
2	CS241312	Object Oriented Programming Lab	0	0	2	1	15	35
3	CS241318	Python Programming Lab	0	0	2	1	15	35
4	SI241321	Social Internship	0	0	4	2	-	50
TOTAL			18	2	10	23	195	605
Total Contact Hours per week: 30								
Total Credits: 23								

N.B. AC241307 is a Mandatory Audit Course (No Credit). It will be evaluated as PP (Pass) or NP (Not Pass)

DETAILED SYLLABUS

Course Code	Course Title	Hours per week L-T-P	Credit C
CS241301	Data Structure and Algorithms	3-0-0	3

Course Outcomes (COs):

- **CO1: Analyze** the time and space complexity of simple algorithms and code segments using asymptotic notations. (*Analyzing – Level 4*) (*Module 1*)
- **CO2: Choose** among the fundamental data structures and **apply** them to solve computational problems. (*Applying – Level 3*) (*Module 2*)
- **CO3: Choose the** efficient searching and sorting algorithms and **apply** them for problem solving (*Applying – Level 3*) (*Module 3*)
- **CO4: Apply** tree and graph for problem solving. (*Applying – Level 3*) (*Modules 4, 5*)
- **CO5: Evaluate** the performance of advanced data structures such as heaps, priority queues, and hash tables for problem solving, enhancing data storage and retrieval efficiency. (*Evaluating – Level 5*) (*Modules 6, 7*)

Module 1: The notion of data structures and algorithms; understanding worst-case, average-case time/space complexity; Basic introduction to asymptotic notations and the growth of functions.

Module 2: Definition of abstract data-type (ADTs); arrays and linked list; Stacks, Queues and their implementations using arrays, linked lists; Doubly linked lists and implementation; prefix, postfix, infix notations and uses of stacks and queues in expression evaluation and conversion.

Module 3: Linear search and Binary Search; Bubble sort, insertion sort, selection sort; divide and conquer paradigm: merge sort and Quick sort with average and worst case analysis, randomized quicksort; Choosing better data structure for different kinds of search and sort

Module 4: Basic definition of graph and tree; Representation of graph: Adjacency list and adjacency matrix; Tree ADT and examples; Implementation of trees and basic traversal algorithms: In- order, pre-order and post-order traversals; Binary trees

Module 5: Binary Search Trees: definition and some basic algorithms; AVL trees: height balance condition, rotations; Introduction to Multi-way search trees, Informal discussion of extension to B-trees.

Module 6: Priority Queue ADT and basic implementation methods; Definition of heaps: Binary heap and binomial heap; Implementation of Priority Queues using heaps and running time analysis; Implementation of heaps using arrays; Heap-sort

Module 7: Hash Tables and hash functions; Collision resolution schemes: chaining, open addressing schemes like linear probing, quadratic probing, double hashing

Text / Reference Books:

1. “Expert Data Structures with C” by R.B. Patel, Khanna Book Publishing Company, New Delhi, 4th Edition.
2. “Fundamentals of Data Structures”, Illustrated Edition by Ellis Horowitz, Sartaj Sahni, Computer Science Press.
3. Introduction to Algorithms – Cormen et al.

Course Code	Course Title	Hours per week L-T-P	Credit C
CS241302	Object Oriented Programming	3-0-0	3

Course Outcomes (COs):

- **CO1: Demonstrate** proficiency in using programming environments, debugging tools, and source code management systems for efficient software development and maintenance. *(Understanding – Level 2) (Module 1)*
- **CO2: Apply** object-oriented programming (OOP) principles to design maintainable and well-structured software solutions. *(Applying – Level 3) (Module 2)*
- **CO3: Implement** advanced OOP features in C++ and Java to develop scalable and efficient applications. *(Applying – Level 3) (Modules 3, 4, 5)*
- **CO4: Develop** robust software using unit testing frameworks and defensive programming techniques to ensure reliability and error handling. *(Creating – Level 6) (Module 6)*
- **CO5: Create** software architectures using software modeling techniques and design patterns such as class diagrams and sequence diagrams. *(Creating – Level 6) (Module 7)*

Module 1: Understanding the build system, IDE, debugging, profiling, and source code management

Module 2: Introduction to Object-Oriented Paradigm: Data encapsulation, modularity, code reuse, identifying classes, attributes, methods and objects, class relationships

Module 3: Interfaces, inheritance, polymorphism, abstract classes, immutability, copying and cloning objects

Module 4: Basics of C++ language, control structures, classes and objects, inheritance

Module 5: Basics of Java language, control structures, classes and objects, inheritance, interfaces, Exception handling

Module 6: Unit testing, developing test suite, Exception handling, assertions

Module 7: Data Structure and Algorithms Laboratory

Text Book / Reference Books:

1. Grady Booch, Robert A. Maksimchuk, Michael W. Engle, Bobbi J. Young, Jim Conallen, Kelli A. Houston. Object-Oriented Analysis and Design with Applications.
2. M. Scott. Programming Language Pragmatics. 4th edition.
3. Object-Oriented Programming with C++, Balaguruswamy, McGraw Hill Education
4. Ivan Bay Ross, Web Enabled Commercial Application using Java 2, BPB publication (1998)
5. R. Sebesta. Concepts of Programming Languages. 10th edition
6. J. Rumbaugh et al. The Unified Modeling Language Reference Manual.
7. Erich Gamma, Richard Helm, Ralph Johnson, John Vlissides, and Grady Booch. Design Patterns: Elements of Reusable Object-Oriented Software.
8. P. Van Roy and S. Haridi. Concepts, Techniques, and Models of Computer Programming.

Course Code	Course Title	Hours per week L-T-P	Credit C
CS241303	Software Engineering	3-0-0	3

Course Outcomes (COs):

- **CO1:** To **select** the most appropriate software process models for a given project scenario. **(Applying – Level 3) (Module 1)**
- **CO2:** **Apply** project estimation and risk management techniques to plan, schedule, and manage software projects efficiently. **(Applying – Level 3) (Module 2)**
- **CO3:** **Develop** a comprehensive Software Requirements Specification (SRS) document based on requirement engineering principles, **Design** efficient software architectures using appropriate design methodologies and **propose** coding styles. **(Creating – Level 6) (Module 3, 4)**
- **CO4:** **Evaluate** software quality through testing techniques and SQA practices to ensure reliability and maintainability of software systems. **(Evaluating – Level 5) (Module 5)**
- **CO5:** **Apply** the modern Computer Aided Software Engineering (CASE) tools in the software life cycle to improve process automation and software quality. **(Applying – Level 3) (Module 6)**

Module 1: Introduction and Software Process Models

Software, Software Engineering, Software Process, Importance of Software Engineering, Standard for Software Process, Waterfall Model, Prototyping Model, Iterative Enhancement Model, Spiral Model, RAD model, 4th Generation models, Formal Methods, Agile Development

Module 2: Software Project Management

Role of Management in Software Development, Project Estimation Techniques, Staffing, Scheduling, Earned Value Analysis, Software Risks, Software Configuration Management, Software Process and Project metrics.

Module 3: Requirement Engineering

Software Requirements, Types of Requirements, Requirement Engineering Cycle, Requirements Specification document, Characteristics of Requirements, Requirement verification and validation.

Module 4: Software Design and Coding

Process, Data and Behavioural Modelling, Design Concepts, Modularity, Architectural design, Coupling and Cohesion, Top-down and bottom-up design, Object-oriented Analysis, Function oriented and Object-Oriented Design approach, Software Design Document, Coding styles and documentation

Module 5: Testing and Software Quality

Testing principles, testing strategies, Black-box and White-box Testing Techniques, Levels of testing -unit, integration, system, regression, Test Plan, Test Cases Specification, Software debugging, Software Maintenance, Software Quality Assurance (SQA). Software Reliability. Software Availability.

Module 6: Computer Aided Software Engineering and Advanced Topics

Computer Aided Software Engineering (CASE) and its Scope, CASE support in Software Life Cycle, Architecture of CASE Environment, Upper CASE and Lower CASE, Exposure to CASE tools. Software Process Improvement, Component Based Software Engineering, Web Engineering, Reverse Engineering, Software Engineering challenges of Big Data, Mobile Applications.

Text / Reference Books:

1. R. S. Pressman, Software Engineering - A practitioner's approach, Ill Edition, McGraw
2. Hill International editions, 1992

3. Ian Sommerville, Software Engineering, Pearson Education Asia, VI Edition, 2000
4. PankajJalote, An Integrated Approach to software Engineering, Springer Verlag, 1997
5. James F. Peters and WitoldPedryez, Software Engineering — An Engineering Approach, John Wiley and Sons, New Delhi

Course Code	Course Title	Hours per week L-T-P	Credit C
CS241304	Digital Systems	3-1-0	4

Course Outcomes (COs):

- **CO1: Demonstrate** familiarities with the Fundamentals of digital systems to analyze and implement basic digital circuits. (**Understanding – Level 2**) (*Module 1*)
- **CO2: Analyze** and construct designing combinational digital circuits (**Analyzing – Level 4**) (*Module 2*)
- **CO3: Analyze** and construct sequential digital circuits. (**Analyzing – Level 4**) (*Module 3*)
- **CO4: Apply** A/D and D/A conversion techniques in digital systems. (**Applying – Level 3**) (*Module 4*)
- **CO5: Design and differentiate** memory architectures and programmable logic devices for efficient digital system design. (**Creating – Level 6**) (*Module 5*)

MODULE 1: Digital signals, digital circuits, AND, OR, NOT, NAND, NOR and Exclusive-OR operations, Boolean algebra, examples of IC gates, number systems-binary, signed binary, octal hexadecimal number, binary arithmetic, one's and two's complements arithmetic, codes, characteristics of digital ICs, digital logic families, TTL, Schottky TTL and CMOS logic, interfacing CMOS and TTL, Tri-state logic. Representation of floating point numbers in IEEE 754 standards.

MODULE 2: Standard representation for logic functions, K-map representation, simplification of logic functions using K-map, minimization of logical functions. Don't care conditions, Multiplexer, De Multiplexer Decoders, Adders, Substractors, BCD arithmetic, carry look ahead adder, serial adder, ALU, elementary ALU design, popular MSI chips, digital comparator, parity checker generator, code converters, priority encoders, decoders drivers for display devices, Q-M method of function realization. Simplification of logic function by algebraic methods.

MODULE 3: A I-bit memory, the circuit properties of Bi stable latch, the clocked SR flip flop, J- K- T and D types flip flops, applications of flip flops, shift registers, applications of shift registers, serial to parallel converter, parallel to serial converter, ring counter, sequence generator, Ripple (Asynchronous) counters, synchronous counters, counters design using flip flops, special counter IC's, asynchronous sequential counters, applications of counters.

MODULE 4: Digital to analog converters: weighted resistor/converter, examples of D/A converter ICs, sample and hold circuit, analog to digital converters: quantization and encoding, parallel comparator A/D converter, counting A/D converter

MODULE 5: Memory organization and operation, expanding memory size, classification and characteristics of memories, sequential memory, read only memory (ROM), read and write memory(RAM), content addressable memory (CAM), charge de coupled device memory (CCD), commonly used memory chips, ROM as a PLD, Programmable logic array, Programmable array logic.

Text / Reference Books:

1. R. P. Jain, "Modern Digital Electronics", McGraw Hill Education, 2009.
2. M. M. Mano, "Digital logic and Computer design", Pearson Education India, 2016.
A. Kumar, "Fundamentals of Digital Circuits", Prentice Hall India, 2016
3. Albert Malvino, Jerald Brown, "Digital Computer Electronics"

Course Code	Course Title	Hours per week L-T-P	Credit C
MA241305D	Mathematics III-D	3-0-0	3

Course Outcomes (COs):

- **CO1:** Students will be able to apply the knowledge of algebraic structures to understand the basics of theoretical foundation of computer science.
- **CO2:** Students will be able to apply the basic principles of Probability and probability distributions in the context of data science.
- **CO3:** Students will be able to analyse and draw conclusions from sample data.
- **CO4:** Students will be able to apply the graph theoretic techniques to solve the problems of Computer Science and Engineering.

Module 1: Definitions and basic properties of Semigroups, Monoids and Groups. Subgroups and Normal Subgroups. Group homomorphism and the related theorems. Basics of Rings, Integral Domains and Fields. Introduction to Lattices and Boolean Algebras. (Contact hours: 8, Marks: 14)

Module 2: Axiomatic definitions of Probability, Conditional Probability. Independence, Rule of total probability, Bay's Rule. Random Variables and their distributions. Cumulative distribution function and its properties. Discrete, Continuous and Mixed random variables. Probability mass and density functions. Mean, Variance and Moments. Characteristic and Moment generating functions. Binomial, Poisson, Uniform and Normal distributions Functions of random variables. Joint distributions. Covariance and Correlations. (Contact hours: 12, Marks: 21)

Module 3: Basics of statistical inference. Sampling and standard error. Point and interval estimations. Maximum likelihood estimator. Level of significance. Test of significance for large and small samples. Student's t-distribution, Chai-square distribution. (Contact hours: 6, Marks: 11)

Module 4: Graphs and their representations by adjacency and incidence matrices. Varieties of graphs. Walks and connectedness. Degrees. Isomorphism of graphs. Cut points, bridges and blocks. Tree and its properties. Binary trees. Eulerian and Hamiltonian graphs. Planar graphs and duals. Thickness and crossing numbers. Colourability of graphs. The five colour theorem. Chromatic number and chromatic polynomial. (Contact hours: 14, Marks: 24)

Textbooks/ References.

1. J. P. Trembly and R. P. Manohar, Discrete Mathematical Structures with application to Computer Science, Tata McGraw Hill, 2001.
2. S. Ross, A first course in Probability, Pearson Education India, 2002.
3. R. K. Jain and S. R. K. Iyengar, Advanced Engineering Mathematics, Narosa, 2016.
4. Kenneth H Rosen, Discrete Mathematics and its Applications, Tata McGraw Hill, 2007.
5. Narsingh Deo, Graph theory with applications to Engineering and Computer Science, Prentice Hall.
6. J. Medhi, Statistical Methods: An introductory text, New Age International Publishers.

Course Code	Course Title	Hours per week L-T-P	Credit C
HS241306	Indian Knowledge Systems	2-0-0	2

Course Objectives:

1. To introduce students to the foundational concepts of Indian Knowledge Systems (IKS).
2. To explore the contributions of ancient and medieval India in science, technology, mathematics, and philosophy.
3. To develop an appreciation for traditional Indian knowledge and its relevance in modern contexts.
4. To encourage critical thinking about indigenous knowledge systems and their applications.

Course Outcome (CO):

After completion of the course, the students will be able to

- **CO1:** Understand the key principles of Indian Knowledge Systems.
- **CO2:** Recognize India's contributions to science, mathematics, medicine, and engineering.
- **CO3:** Analyze the interdisciplinary nature of IKS and its connections with modern technology.
- **CO4:** Appreciate the philosophical and ethical dimensions of traditional Indian knowledge.

MODULE 1: Introduction to Indian Knowledge System (IKS) (4 Lectures)

- Definition, scope, and significance of IKS
- Sources of Indian knowledge: Vedas, Upanishads, Puranas, Smritis, and other texts
- Classification of knowledge in Indian tradition (Vidyas and Kalas)

MODULE 2: Science & Technology in Ancient India (12 Lectures)

- Contributions to mathematics (Zero, Decimal System, Vedic Mathematics, Aryabhata, Brahmagupta)
- Astronomy & Space Science (Surya Siddhanta, Planetary Motion, Concept of Time)
- Metallurgy and material sciences (Iron Pillar, Wootz Steel, Ayurvedic alloys)
- Civil engineering & architecture (Temple architecture, Town planning, Water management)

MODULE 3: Indian Philosophical & Ethical Systems (6 Lectures)

- Key schools of Indian philosophy (Nyaya, Vaisheshika, Samkhya, Yoga, Vedanta)
- Concept of Dharma, Karma, and sustainable living
- Ethics in science and technology from an Indian perspective

MODULE 4: Traditional Medicine & Wellness Systems (4 Lectures)

- Foundations of Ayurveda, Yoga, and Siddha
- Holistic health and wellness practices
- Relevance of traditional medicine in modern healthcare

MODULE 5: IKS in Contemporary Applications (4 Lectures)

- Revival of traditional knowledge in modern science and engineering
- Case studies (e.g., Panchakarma in wellness tech, Vastu in architecture)
- Government initiatives (National Education Policy 2020, IKS Division under AICTE)

Text Books:

1. "Indian Knowledge Systems" – Kapil Kapoor & Avadhesh Kumar Singh
2. "Ancient Indian Leaps into Mathematics" – B.S. Yadav
3. "Science and Technology in Ancient India" – O.P. Jaggi
4. "The Positive Sciences of the Ancient Hindus" – Brajendranath Seal
5. AICTE Model Curriculum for IKS (Available online)

Online Resources:

- NPTEL / SWAYAM courses on IKS
- AICTE-IKS Portal (<https://iksindia.org/>)
- Digital Library of India (<https://dli.gov.in>)

Course Code	Course Title	Hours per week L-T-P	Credit C
AC241307	Language other than MIL	1-1-0	0

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Demonstrate a clear understanding of basic grammar rules including subject-verb agreement, tenses, and cases, and apply them in constructing grammatically correct sentences.
- **CO2:** Identify and correct common grammatical and usage errors in written English, enhancing sentence effectiveness and clarity.
- **CO3:** Enhance their vocabulary through the use of synonyms, antonyms, idiomatic expressions, and phrasal verbs, improving both written and spoken communication.
- **CO4:** Apply verbal reasoning skills to comprehend, arrange, and complete sentences and paragraphs effectively.
- **CO5:** Develop strong reading comprehension skills using skimming, scanning, and intensive reading techniques to extract key information from various texts.
- **CO6:** Produce coherent and concise writing through precis writing and essay composition, demonstrating logical flow, clarity, and appropriate structure.

MODULE 1: Basic grammar usage

(10 Lectures)

Subject-verb agreement, narration, tenses

Case: Subjective case, possessive case, objective case

Writing effective sentences, Spotting errors

MODULE 2: Vocabulary Building

(6 Lectures)

Synonyms and Antonyms, One word substitution,

Idioms and Phrases and Phrasal Verbs

MODULE 3: Verbal ability

(8 Lectures)

Cloze test

Sentence rearrangement, Sentence completion,

Paragraph Jumbles

MODULE 4: Writing and Reading Ability

(8 Lectures)

Reading subskills, Skimming, Scanning and Intensive reading

Reading Comprehension, Precis Writing, Essay Writing

Text Books

1. Objective General English: S.P.Bakshi, Arihant Publication India Ltd., 2021
2. Descriptive English: S.P.Bakshi, Arihant Publication India Ltd., 2016

Course Code	Course Title	Hours per week L-T-P	Credit C
CS241311	Data Structure and Algorithms Lab	0-0-2	1

CO1: Analyze runtime of simple algorithms or code segment. (*Analyzing – Level 4*)

CO2: Implement linear and non-linear data structures such as arrays, stacks, queues, and linked lists to solve computational problems. (*Applying – Level 3*)

CO3: Implement searching and sorting algorithms in terms of time and space complexity. (*Applying – Level 3*)

CO4: Implement tree and graph and perform basic tree and graph traversals using suitable programs. (*Applying – Level 3*)

CO5: Choose and implement appropriate data structures for specific application scenarios. (*Evaluating – Level 5*)

LIST OF EXPERIMENTS

1. Worst/average case analysis and asymptotic notations for small pseudo-codes
2. Implementation of arrays
3. Implementation of Linear search using array
4. Implementation of Binary Search using array
5. Implementation different kinds of linked list with all possible operations
6. Stacks, Queues implementations using linked list and array
7. Implementation of Bubble sort, insertion sort, selection sort; Merge sort and divide and conquer paradigm; Quick sort using array. Also check possibility of implementing using linked list
8. Implementation of stacks with application to a problem.
9. Implementation of queues with application to a problem.
10. Implementation of binary search trees and basic tree traversal algorithms
11. Implementation of Binary heap;
12. Graph implementation using adjacency list and adjacency matrix
13. Implementation of basic concepts of hashing

Additional Experiments: Additional experiments (if any) considered appropriate by the department may be conducted.

Text Book / Reference Books:

1. “Expert Data Structures with C” by R.B. Patel, Khanna Book Publishing Company, New Delhi, 4th Edition.
2. “Fundamentals of Data Structures”, Illustrated Edition by Ellis Horowitz, Sartaj Sahni, Computer Science Press.
3. “Data Structure Through C” by Yashavant P. Kanetkar, BPB Publications

Course Code	Course Title	Hours per week L-T-P	Credit C
CS241312	Object Oriented Programming Lab	0-0-2	1

- **CO1:** Implement classes and objects to encapsulate data and methods in C++ programs. *(Applying – Level 3)*
- **CO2:** Implement C++ programs using object-oriented concepts such as inheritance, function overloading, operator overloading etc. *(Applying – Level 3)*
- **CO3:** Implement programs that demonstrate the use of constructors, destructors, and access specifiers for effective memory and access control. *(Applying – Level 3)*
- **CO4:** Design and develop object-oriented solutions for real-life scenarios such as library management, banking systems etc. *(Creating – Level 6)*

LIST OF EXPERIMENTS

Experiments: Experiments are to be done with C++ programming language

1. Implementation of a class with data members and member functions
2. Implementation of class with a main program to test create many objects of a class and to use the data members and member functions
3. Implementation of class with a main program to test create array of objects of a class and using the data members and member functions
4. Implement concepts of different types of inheritance
5. Implement the concept of function overloading
6. Implement the concept of Constructors and Destructors
7. Implement class to make use of operator overloading
8. Implementations of classes and object for some real problems like book shops, library, banks etc.
9. Implement applications to learn about access specifiers

Additional Experiments: Additional experiments (if any) considered appropriate by the department may be conducted.

Course Code	Course Title	Hours per week L-T-P	Credit C
CS241318	Python Programming Lab	0-0-2	1

Course Outcomes

- **CO1:** Apply basic building blocks of python programming.
- **CO2:** Develop Python programs using control structures, functions, and data structures to solve real-world problems.
- **CO3:** Implement basics concepts of object oriented programming using Python programs.
- **CO4:** Design and develop interactive graphical user interfaces (GUIs) using Python GUI libraries to create applications and simple games

MODULE 1: Introduction to importance of IDEs like Spyder (Anaconda)/PyCharm for professional programming, explore Python shell as a calculator and for inputting Python expressions directly, HelloWorld program in Python script, Python keyword and Identifiers, Indentation, Comments, Data Types in. Operators in Python: comparison, arithmetic, logical, Boolean, bitwise, assignment. Python: numbers, list, tuple, strings, set, dictionary, conversion between various data types

MODULE 2: Input and Output in Python, if-else , for loop, while loop, break, pass, continue, creating Functions, functions with arguments, returning values from functions, lambda expressions, recursion, global and local variables, Importing other modules/packages and using their functions, creating random numbers/random-choice to create programs for simple guessing games like Rock – Paper-Scissors. Problems on 1D/2D/3D arrays using list. Problem solving using dictionary as look-up table.

MODULE 3: Basics of Object-oriented programming: Class and Object. Defining variables and functions inside class. Creating objects, Inheritance, Multiple and Multi Level Inheritance, Function over-riding, the concept of composing objects of a different class in an object, problems on object composition

MODULE 4: GUI creation using Python’s de-facto GUI package like tkinter or alternative packages like: wxPython, PyQt (PySide), Pygame, Pyglet, and PyGTK. Creating labels, buttons, entry (textbox), combobox, checkbutton, radiobutton, scrolledText (textarea), spinbox, progressbar, menubar, filedialog, tabs etc. Creating GUI simple games like Tic-Tac-Toe

Text Book / Reference Books:

1. Allen B Downey , Think Python 2nd Edition - How to Think Like a Computer Scientist
2. O’Reilly publication, 2nd edition 2016
3. Zed A. Shaw, Learn Python 3 the Hard Way Addison-Wesley, 3rd edition 2017
4. Paul Barry David Griffiths Barry Griffiths Head First Programming: A Learner’s Guide to Programming using the Python Language, O’Reilly publication, 1st edition 2009
5. Mark Pilgrim, Dive into Python 3, Apress publication, 2nd edition 2009

Experiments: Experiments are to be conducted with python programming language. A set of experiments considered appropriate by the department may be conducted for each module.
